Project Support Session

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Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Submission Guidelines
- FAQs

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Project Problem Statement

Business Context

Brain tumors are abnormal growths of cells within the brain and can significantly impact neurological function and overall health. Depending on the tumor type and location, patients may experience symptoms such as headaches, seizures, vision problems, or cognitive difficulties. Accurate identification of brain tumors and differentiation between tumor types is critical for timely diagnosis, treatment planning, and clinical decision-making.

Magnetic Resonance Imaging (MRI) is widely used to examine brain structures and detect abnormalities due to its high-resolution and detailed soft-tissue contrast. However, MRI scans of different brain tumor types-such as glioma, meningioma, and pituitary tumors-can exhibit visually similar patterns. In some cases, tumor images may also resemble healthy brain scans, making manual interpretation challenging and prone to variability across clinicians. Misclassification or delayed diagnosis can affect treatment outcomes and patient prognosis.

Therefore, there is a strong need for a systematic and reliable approach to analyze brain MRI images that can accurately distinguish between different types of brain tumors and normal (healthy) brain conditions.

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Project Problem Statement

Objective

The objectives of this project are to:

- Analyze brain MRI images to identify distinguishing characteristics associated with different brain tumor types and healthy brain tissue.
- Build neural network-based classifiers that can learn meaningful patterns from MRI image data for accurate brain tumor classification.
- Improve the performance and reliability of the classifier through model enhancements, optimization techniques, and better feature learning.

The end goal is to support consistent and reliable assessment of brain conditions using MRI data to aid clinical diagnosis and decision-making.

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Dataset Overview

- The dataset consists of brain MRI images organized into four classes, which are to be converted into numerical arrays for model training and evaluation.

Glioma Tumor [645 images]

MRI scans of patients diagnosed with glioma tumors. Gliomas are one of the most common and aggressive brain tumors, and identifying them separately is critical because their treatment approach differs significantly from other tumor types.

Meningioma Tumor [660 images]

MRI scans representing meningioma tumors, typically arising from the meninges. Meningiomas are often slow-growing and may require different clinical decisions, making it important to distinguish them rather than grouping all tumors into a single category.

Pituitary Tumor [750 images]

MRI images showing tumors located in the pituitary gland region. Pituitary tumors affect hormone regulation and are located in a unique brain region, so recognizing them as a separate class supports more targeted diagnosis and management.

No Tumor [840 images]

MRI scans of healthy individuals with no visible brain tumor. Including healthy scans helps the model learn the difference between normal brain structures and abnormal tumor patterns, reducing false positives.

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Open Q&A

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Project Evaluation Rubric

Section	Points
Requirements Gathering	4
Data Overview	4
Exploratory Data Analysis	4
Data Preprocessing	6
Model Building - Simple Artificial Neural Networks	5
Model Building - Optimized Artificial Neural Networks	5
Model Building - Convolutional Neural Networks (VGG16 Base Model)	5
Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers)	5
Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers) + Data Augmentation	5
Model Performance Comparison and Final Model Selection	5
Business Insights & Recommendations	4
Presentation / Notebook - Overall Quality	8

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Project Evaluation Rubric

Requirements Gathering	2	3	4	5	6	7	8	9	10	11	12

What to Cover?	Guiding Questions and Considerations
Business Problem Definition	Do your problem statement and analysis questions clearly align with the business objective, use data effectively, and aim to generate actionable insights?
Elaborate on the provided requirements	Are the requirements broken down into measurable, data-driven steps that can be validated through analysis?

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Project Evaluation Rubric

1	Data Overview	3	4	5	6	7	8	9	10	11	12

What to Cover?	Guiding Questions and Considerations
Load the dataset	Did the dataset load and extract correctly and completely?
Check the shape of the dataset	Did the number of images from each class match the data description?
Plot random images from each class and print their corresponding labels	What do you observe from the images plotted?

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Project Evaluation Rubric

1	2	Exploratory Data Analysis	4	5	6	7	8	9	10	11	12

What to Cover?	Guiding Questions and Considerations
Image Dimension Distribution	Does any class of images have a different distribution of height and width compared to other classes?
Pixel Value Distribution (Mean of RGB Channels)	Does the RGB intensity distribution differ between classes?

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Project Evaluation Rubric

1	2	3	Data Preprocessing	5	6	7	8	9	10	11	12

What to Cover?	Guiding Questions and Considerations
Resize images	How do we choose the target image size so that models can process the images efficiently?
Prepare the data for modeling	How do we decide the ratio of train-test split? Is data leakage taken care of?
Data Normalization	How do we normalize image data in a way that is appropriate for the chosen model architecture?
Encode the labels	Does the final shape of the label vector match the expected number of nodes in the model's output layer?

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Project Evaluation Rubric

1	2	3	4	Model Building - Simple Artificial Neural Networks	6	7	8	9	10	11	12

What to Cover?	Guiding Questions and Considerations
Choose the metric of choice with proper rationale	Explain the choice of key metric(s) in the context of the business trade-off between actual tumours and false alerts.
Build and compile the model	Does the architecture include an appropriate number of hidden layers and neurons for the complexity of the data?
Train the model	Are the number of epochs and the batch size tuned to ensure the model converges without overfitting?
Check and comment on the performance of the model	How does the model perform on the validation set compared to the training set regarding the metric of choice?

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Project Evaluation Rubric

1	2	3	4	5	Model Building - Optimized Artificial Neural Networks	7	8	9	10	11	12

What to Cover?	Guiding Questions and Considerations
List down the steps to be taken to optimize the model performance with proper rationale	How do we identify the main levers for optimization based on the performance of the Simple ANN?
Build and compile the model	Does the architecture include an appropriate number of hidden layers and neurons for the complexity of the data?
Train the model	Are the number of epochs and the batch size tuned to ensure the model converges without overfitting?
Check and comment on the performance of the model	How does the model perform on the validation set compared to the training set regarding the metric of choice?

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Project Evaluation Rubric

1	2	3	4	5	6	Model Building - Convolutional Neural Networks (VGG16 Base Model)	8	9	10	11	12

What to Cover?	Guiding Questions and Considerations
Build and compile the model	How do we choose which layers of VGG16 to freeze? How do we design the classifier head on top of VGG16?
Train the model	Are the number of epochs and the batch size tuned to ensure the model converges without overfitting?
Check and comment on the performance of the model	How does the model perform on the validation set compared to the training set regarding the metric of choice?

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Project Evaluation Rubric

1	2	3	4	5	6	7	Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers)	9	10	11	12

What to Cover?	Guiding Questions and Considerations
Build and compile the model	Are the Feedforward Layers (Dense layers) appropriately structured to transition from the flattened convolutional output to the final classification?
Train the model	Are the number of epochs and the batch size tuned to ensure the model converges without overfitting?
Check and comment on the performance of the model	How does the model perform on the validation set compared to the training set regarding the metric of choice?

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Project Evaluation Rubric

1	2	3	4	5	6	7	8	Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers) + Data Augmentation	10	11	12

What to Cover?	Guiding Questions and Considerations
Implement data augmentation	How do we choose which augmentation operations are suitable for the domain and business problem at hand?
Build and compile the model	Are the Feedforward Layers (Dense layers) appropriately structured to transition from the flattened convolutional output to the final classification?
Train the model	Are the number of epochs and the batch size tuned to ensure the model converges without overfitting?
Check and comment on the performance of the model	How does the model perform on the validation set compared to the training set regarding the metric of choice?

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Project Evaluation Rubric

1	2	3	4	5	6	7	8	9	Model Performance Comparison and Final Model Selection	11	12

What to Cover?	Guiding Questions and Considerations
Compare the model performances of all models on both training and validation sets	Are model performances compared consistently across training and validation sets? Do the results explicitly address overfitting or underfitting?
Choose the best model with proper rationale	Business end goal drives the best model choice
Check and comment on the performance of the best model on the test set	Does the model performance generalize well to the unseen test set? Is there a significant drop in performance compared to the validation results?

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Project Evaluation Rubric

1	2	3	4	5	6	7	8	9	10	Business Insights & Recommendations	12

What to Cover?	Guiding Questions and Considerations
Conclude with the key takeaways for the business	Are the key insights from modeling included in final takeaways, along with comments on their alignment to the business objective? Logical next steps are a key inclusion

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Project Evaluation Rubric

1	2	3	4	5	6	7	8	9	10	11	Presentation / Notebook - Overall Quality

Low-Code Considerations	Full-Code Considerations
Structure and flow - everything sits well in a story	Clear structure and flow
Crispness - focus on key points	Well-commented and executed code
Visual appeal - layout supports readability	Conclusion and Business Recommendations
Conclusion and Business Recommendations	

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Project Submission Guidelines: Low-Code

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Template Notebook: Low-Code Version* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

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Project Submission Guidelines: Low-Code

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

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Project Submission Guidelines: Full-Code

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution from scratch

Download the dataset and the *Template Notebook: Full-Code Version* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Clearly write down observations, insights, and recommendations for the business problem based on the analysis

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

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Project Submission Guidelines: Full-Code

The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)

The conversion can be done via **free online tools**

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

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FAQs

How should I choose the final model when comparing multiple trained versions?

Primary metric alignment

Select the model that optimises the metric that actually reflects the business objective, for example recall for minority classes or precision where false positives are costly. Accuracy alone is usually misleading with class imbalance.

Generalisation gap

Compare validation and test performance. Large gaps indicate overfitting, often seen in aggressively tuned models. A slightly weaker but stable model is usually the safer choice.

Model complexity and robustness

Hyperparameter tuning increases variance and operational risk. If gains over the base model are marginal, the simpler model is often the better engineering decision.

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FAQs

Why must we freeze the VGG16 base layers during initial training?

The VGG16 base **has already learned complex patterns** (edges, textures, shapes) from millions of images in the ImageNet dataset.

If we don't freeze these layers, the randomly initialized weights of your new feedforward layers will backpropagate large gradients into the base, **destroying these useful pre-trained features**.

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FAQs

When should I stop training the model?

We should monitor the **Learning Curves**.

If the **training loss continues to go down while the validation loss starts to climb**, the model is **overfitting**.

Implementing **EarlyStopping** is a best practice to halt training at the point where validation performance is at its peak.

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FAQs

How does Data Augmentation help if the images are already normalized?

Normalization (scaling pixels to 0-1) **helps the model converge faster** by keeping gradients stable.

Data Augmentation addresses **overfitting**.

By creating "new" versions of images through rotation or flipping, you **force the model to learn the actual object** rather than memorizing the specific orientation or position of pixels in the training set.

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Power Ahead!

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